CS106.3

Algorithms and Data Structures

Lab 1

Structures

1. Write down the difference between an array and the structure?

2. How many bits will be allocated in the memory when we creating the following structure?

```
struct Book {
   char  title[50];
   char  author[50];
   char  subject[100];
   int   book_id;
}harrypotter;
```

3. Declare above structure as a local structure and initialize values directly?

4. What is the meaning of the last line of the above structure?

5. What will happen if we add typedef to the above structure?

```
typedef struct Book {
   char  title[50];
   char  author[50];
   char  subject[100];
   int   book_id;
}harrypotter;
```

6. What will be the output of the below program?

```
1. #include<stdio.h>
2. main()
3. {
```

```
4.    struct abc {
5.      int a;
6.      char cha;
7.    }var;
8.    var.cha = 'A';
9.    printf("%d", var.a);
10.}
```